3MB3 Final Project

Arms Race

Zeyn Jaswal, Azzaam Khan, Timothy Palin, Meredith Reeves, and Tony Xu

McMaster University

February 25, 2026

Arms Race Model

Model Equations

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$

$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

- a and c: Mutual Fear Factor ($a, c > 0$)
- b and d: Expenditure Factor ($b, d > 0$)
- r and s: Goodwill/Grievance Factor (($r, s < 0$) for goodwill, ($r, s > 0$) for grievance)

Stability Analysis

Stability Equilibria

$$p^*(t) = \frac{as + rd}{bd - ac}$$

$$g^*(t) = \frac{cr + sb}{bd - ac}$$

Stability Condition: $bd > ac$

Long Term Behavior

Mutual Logistic Fear

End

Thank you for listening

[0] Presentation theme by Ricardo B Lourenco